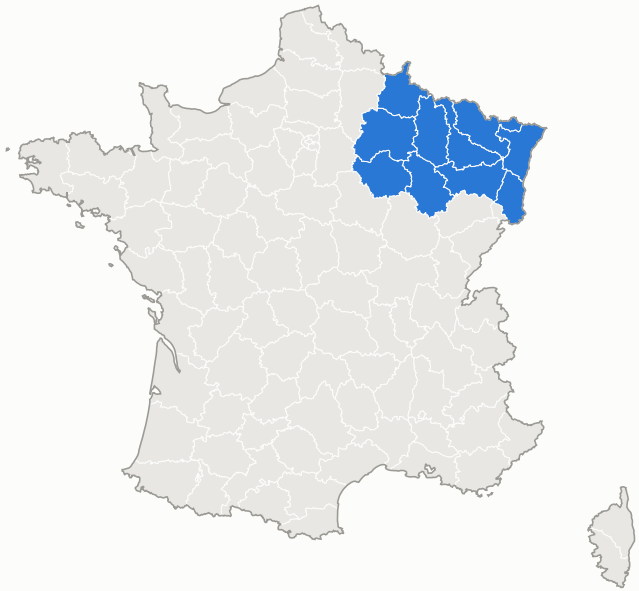
Forest resources of Grand Est (France), 2023

National forest inventory indicators, generated by ForestDataExplorerExample.jl on 2026-10-22

Location



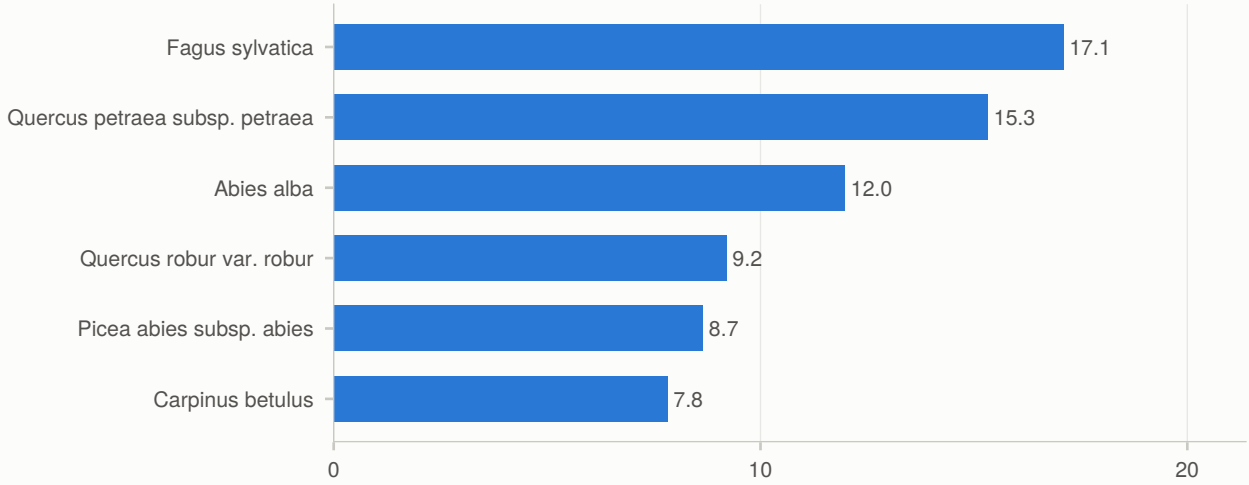
Key figures (all species)

Indicator	Unit	2023	2022	Mean 2018–2022	Variability
Growing stock	million m³	**402**	376	381	3.6 %
Forest area	million ha	**1.81**	1.81	1.81	0.0 %
Growing stock per hectare	m³/ha	**222**	208	211	3.6 %
Biological growth	million m³/yr	**11.0**	10.4	11.0	4.0 %
Harvest	million m³/yr	**9.69**	8.25	8.52	14.5 % ▲
Natural mortality	million m³/yr	**3.47**	3.20	2.65	15.4 % ▲
Mortality rate	%/yr	**0.86**	0.85	0.70	15.9 % ▲
Harvest / growth	ratio	**0.88**	0.80	0.78	11.3 % ▲

Variability: mean absolute year-on-year change relative to the mean level over 2005–2025. ▲ marks the variables above 5 %, compared with the previous years in this report.
Harvest and mortality are measured on plots revisited after five years (from 2015). Areas come from the land cover map; see the methodology page.

Main species

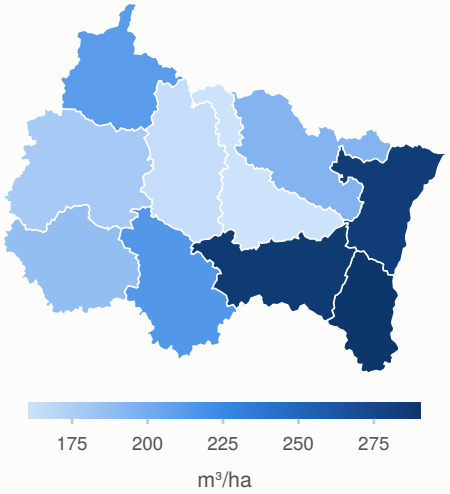
share of the growing stock in 2023, %



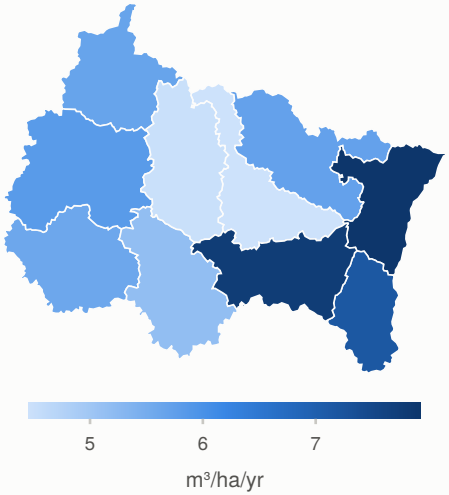
Indicators by department, 2023

All species. Grey: no estimate.

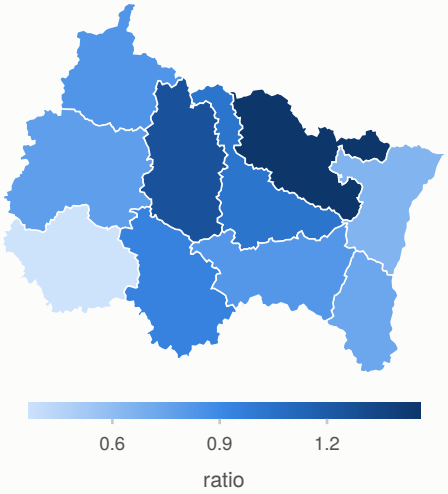
Growing stock per hectare



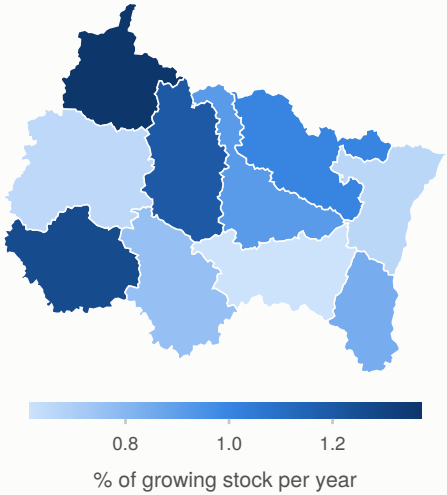
Biological growth per hectare



Harvest / biological growth



Mortality rate



Department	Stock (m³/ha)	Growth (m³/ha/yr)	Harvest / growth	Mortality (%/yr)
Ardennes	212	5.68	0.83	1.37
Aube	187	5.61	0.36	1.26
Bas-Rhin	284	7.93	0.65	0.66
Haut-Rhin	291	7.17	0.73	0.84
Haute-Marne	215	5.18	0.94	0.76

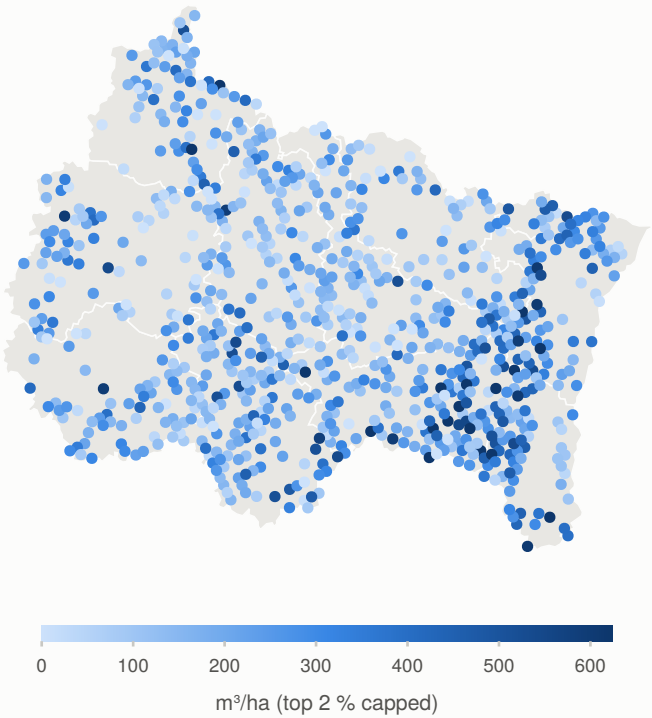
Department	Stock (m³/ha)	Growth (m³/ha/yr)	Harvest / growth	Mortality (%/yr)
Marne	179	5.83	0.78	0.66
Meurthe-et-Moselle	161	4.46	1.03	0.91
Meuse	164	4.51	1.26	1.20
Moselle	194	5.72	1.46	1.00
Vosges	287	7.78	0.82	0.61

Inventory plots and forest cover

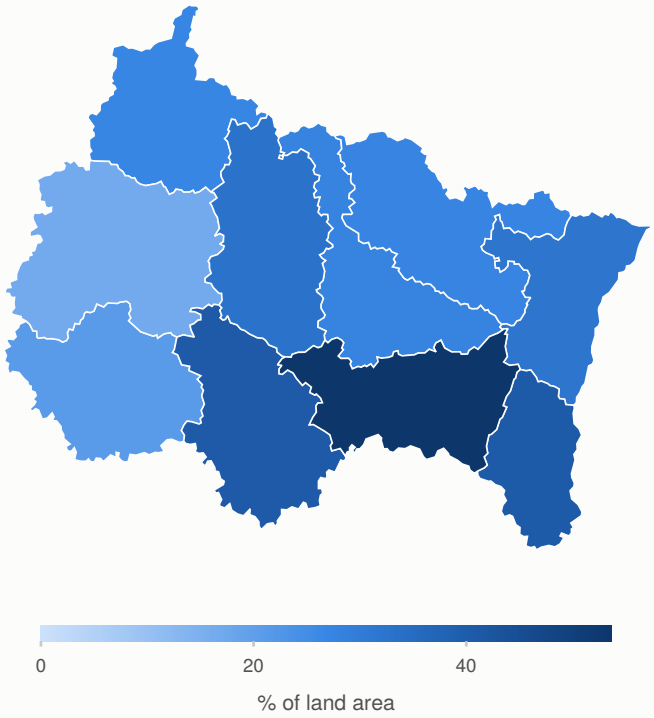
Left: growing stock per hectare measured on the inventory plots of 2023 (all species). Right: forest cover from the land cover map.

Inventory plots, 2023

826 forest plots

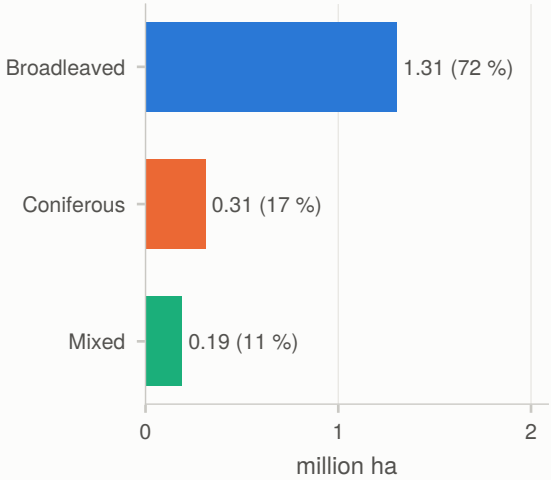


Forest cover by department



Forest area by type

Grand Est

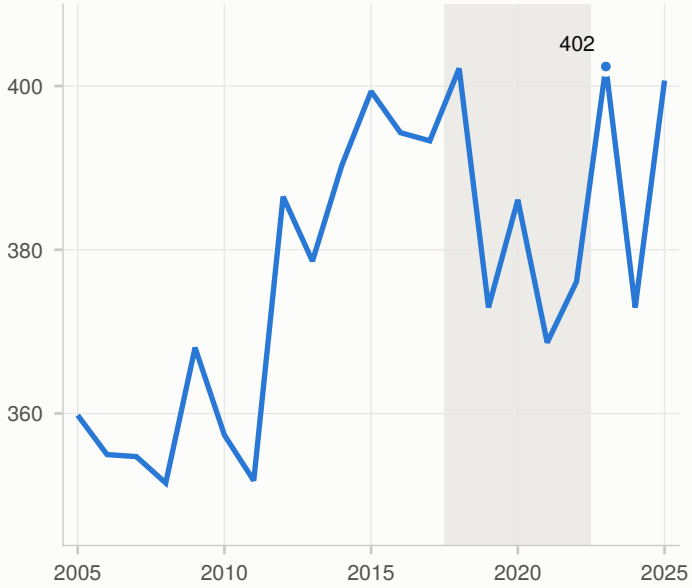


Trends 2005–2025

All species. The shaded band marks the five years before 2023; the dot marks 2023.

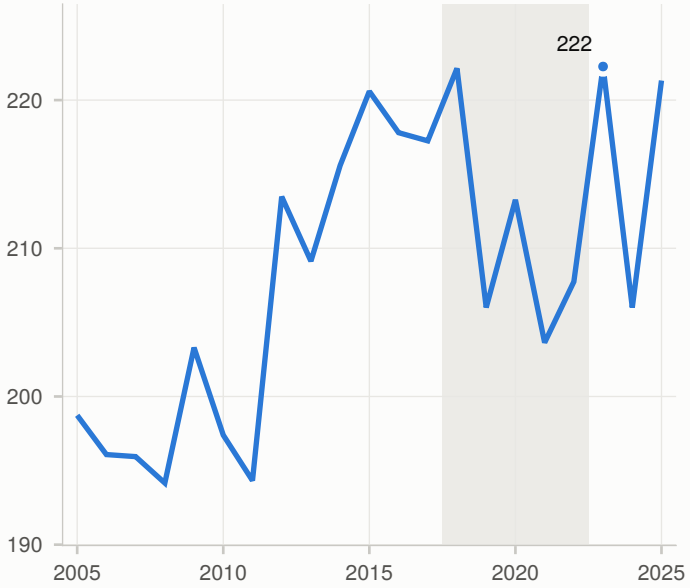
Growing stock

million m³



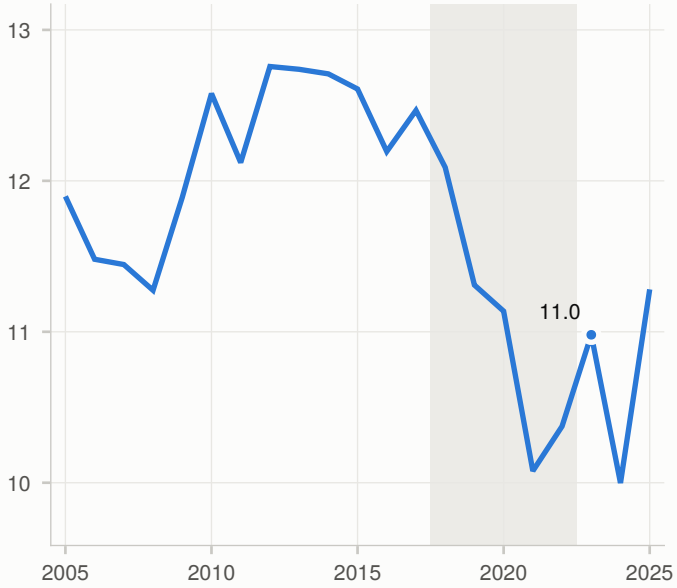
Growing stock per hectare

m³/ha



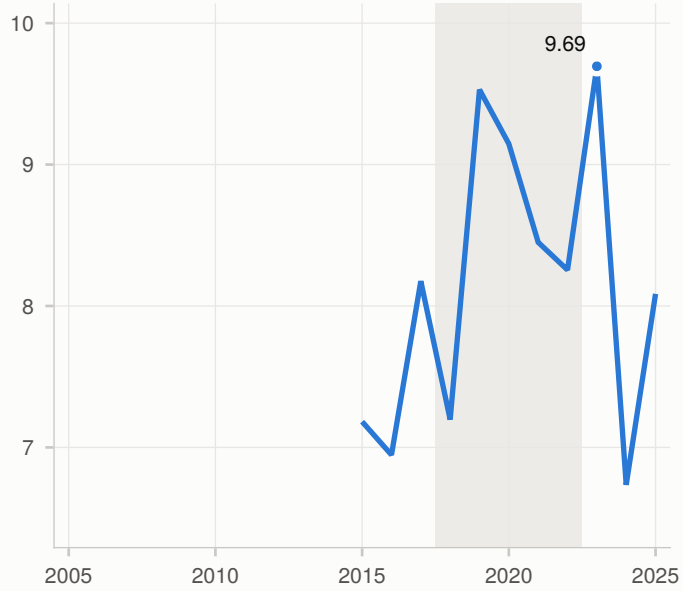
Biological growth

million m³/yr



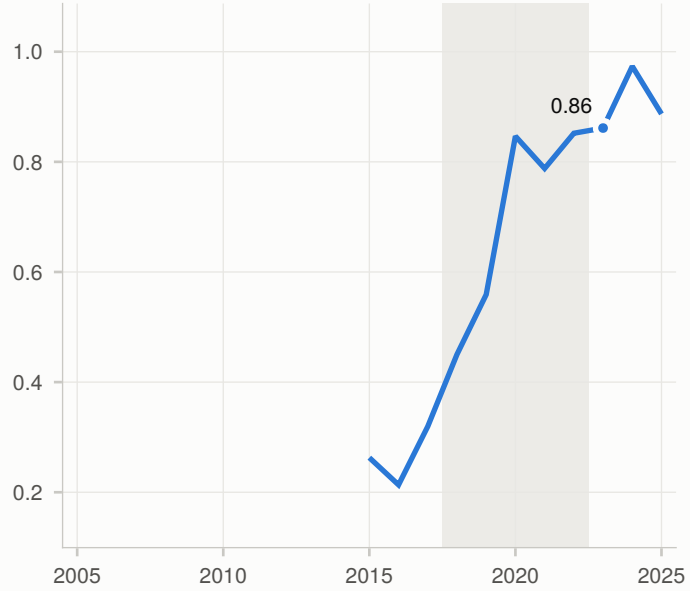
Harvest

million m³/yr



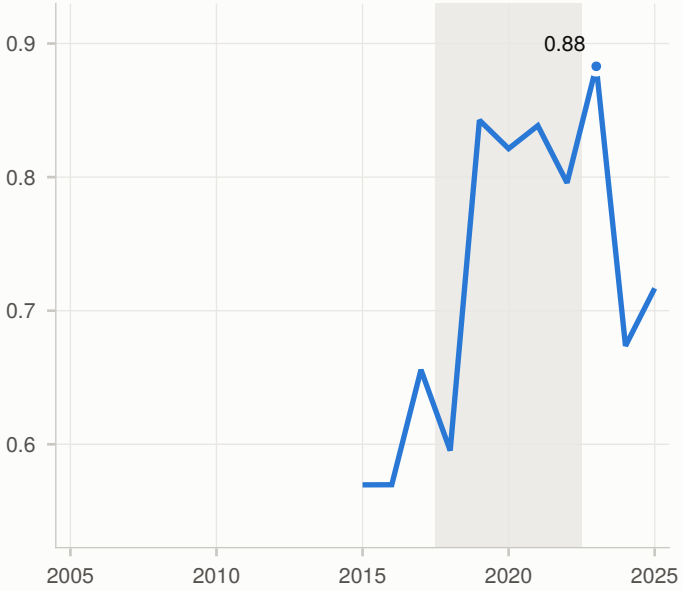
Mortality rate

%/yr



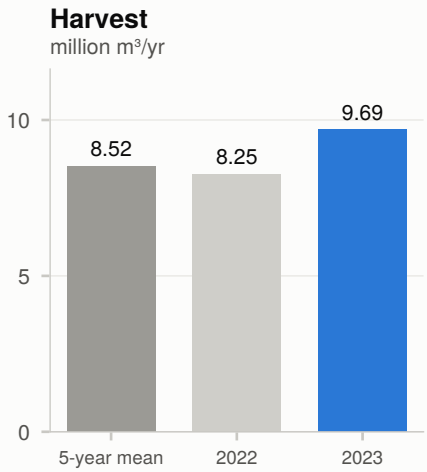
Harvest / growth

ratio

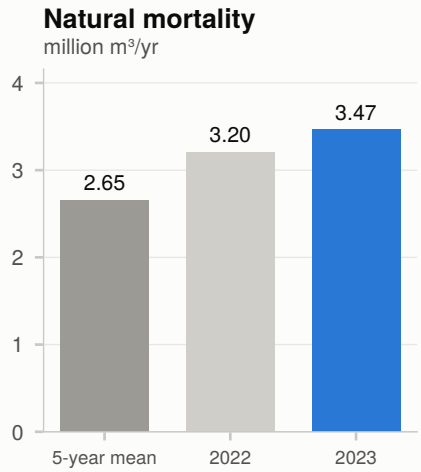


2023 compared with the previous year and the previous five years

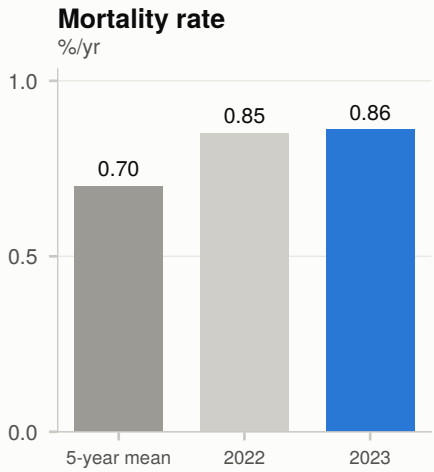
Variables with high temporal variability (mean absolute year-on-year change above 5 %), all species. 5-year mean: 2018–2022.



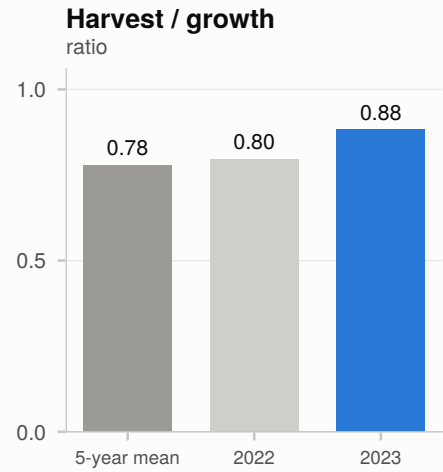
+13.8 % vs 5-year mean



+30.6 % vs 5-year mean



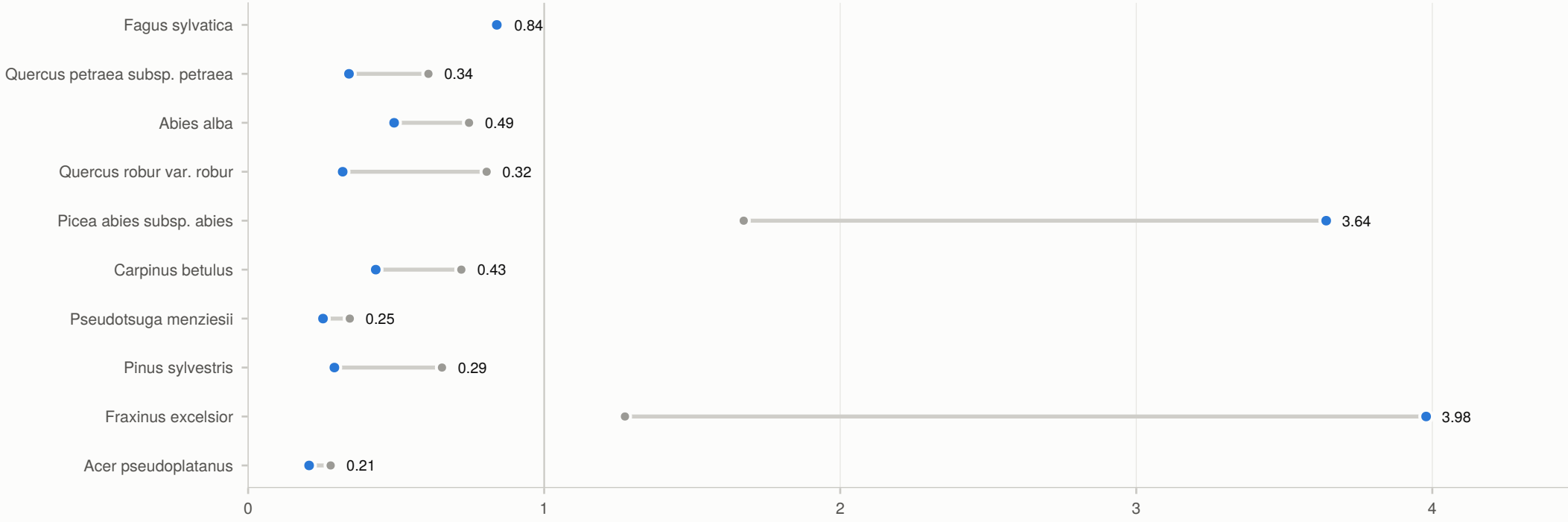
+23.2 % vs 5-year mean



+13.4 % vs 5-year mean

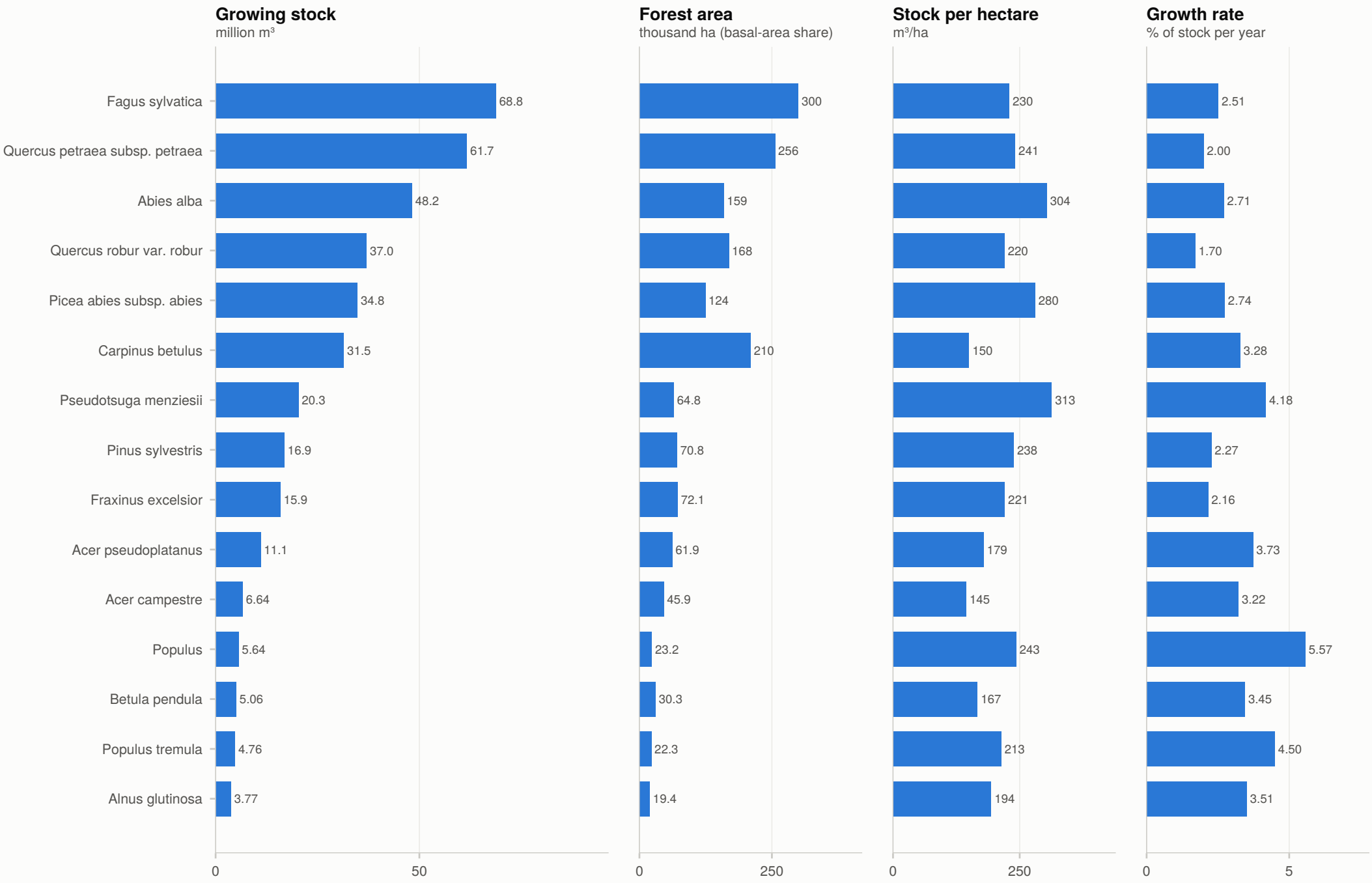
Harvest / biological growth of the main species

Grey: mean 2018–2022; blue: 2023. A ratio above 1 means more volume harvested than grown.



Species composition, 2023

The 15 species with the largest growing stock.

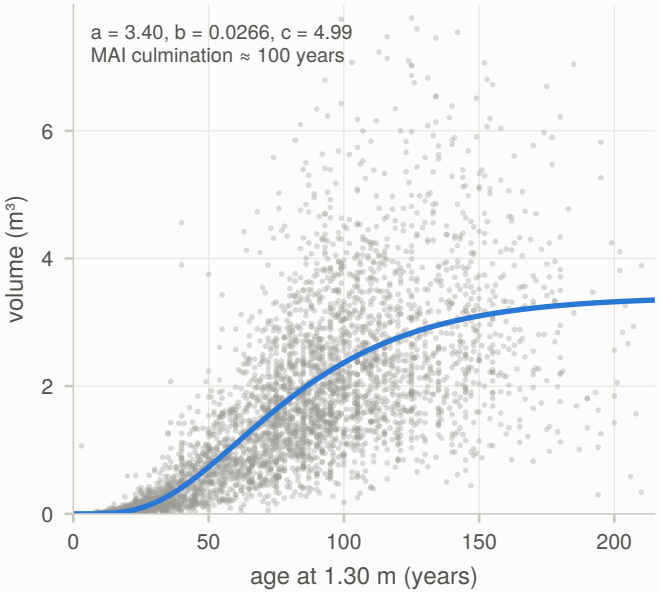


Tree growth models of the main species

Chapman-Richards fit of tree volume on age at 1.30 m, V = a(1 − e^(−b·age))^c, on the trees cored by the inventory (all years). Grey dots: observations; blue line: fitted model.

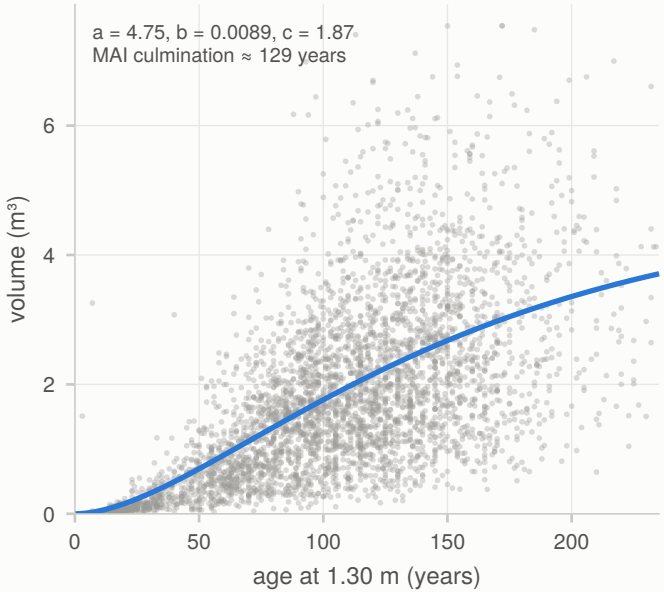
Fagus sylvatica

Grand Est · n = 5206 · R^2 = 0.50



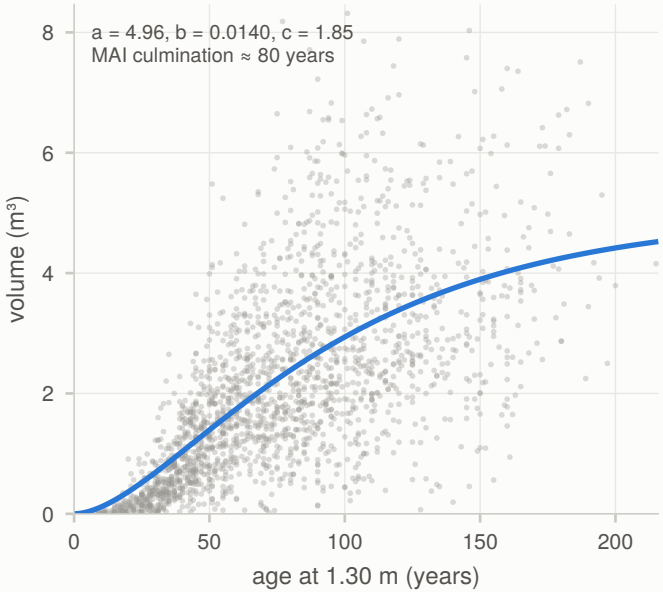
Quercus petraea subsp. petraea

Grand Est · n = 4759 · R^2 = 0.38



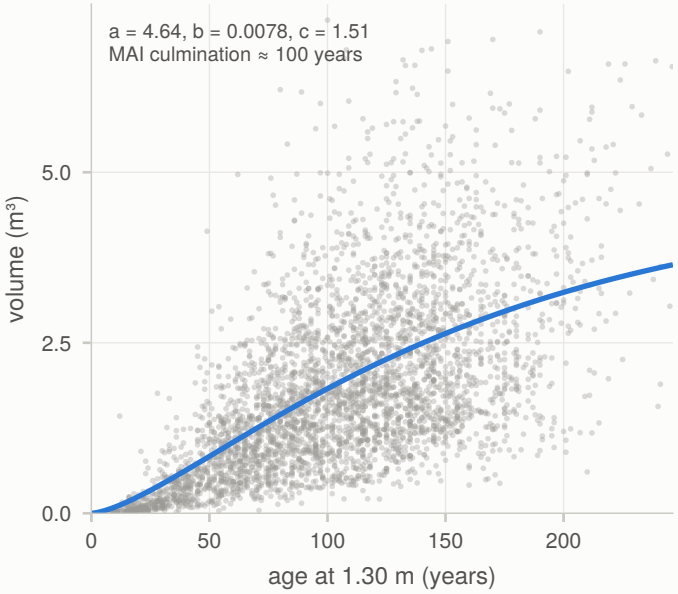
Abies alba

Grand Est · n = 2035 · R^2 = 0.43



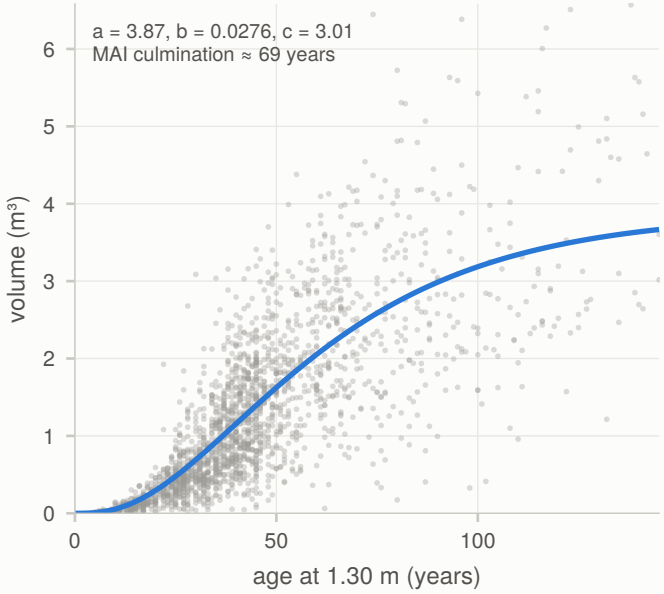
Quercus robur var. robur

Grand Est · n = 3944 · R^2 = 0.35



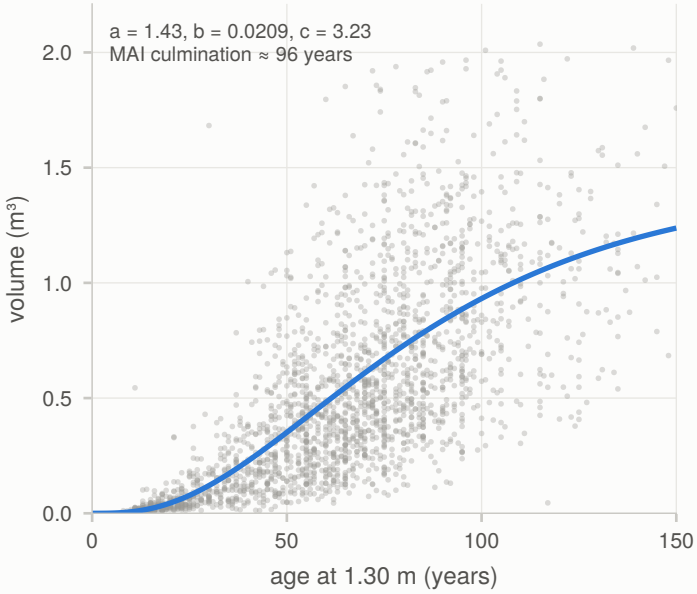
Picea abies subsp. abies

Grand Est · n = 2026 · R^2 = 0.59



Carpinus betulus

Grand Est · n = 2201 · R^2 = 0.46



Indicators by species, 2023

The 22 species with the largest growing stock. – : not estimable.

Species	Growing stock (million m³)	Area (thousand ha)	Stock/ha (m³/ha)	Growth (million m³/yr)	Harvest (million m³/yr)	Mortality (%/yr)	Harvest / growth	Plots (n)
Fagus sylvatica	68.8	300	230	1.73	1.45	0.43	0.84	328
Quercus petraea subsp. petraea	61.7	256	241	1.23	0.42	0.28	0.34	266
Abies alba	48.2	159	304	1.30	0.64	0.82	0.49	103
Quercus robur var. robur	37.0	168	220	0.63	0.20	0.39	0.32	195
Picea abies subsp. abies	34.8	124	280	0.95	3.47	1.91	3.64	99
Carpinus betulus	31.5	210	150	1.03	0.45	0.83	0.43	312
Pseudotsuga menziesii	20.3	64.8	313	0.85	0.21	0.39	0.25	37
Pinus sylvestris	16.9	70.8	238	0.38	0.11	0.46	0.29	64
Fraxinus excelsior	15.9	72.1	221	0.34	1.37	3.83	3.98	115
Acer pseudoplatanus	11.1	61.9	179	0.41	0.09	0.12	0.21	111
Acer campestre	6.64	45.9	145	0.21	0.04	0.27	0.18	105
Populus	5.64	23.2	243	0.31	0.68	1.73	2.15	15
Betula pendula	5.06	30.3	167	0.17	0.03	3.49	0.17	89
Populus tremula	4.76	22.3	213	0.21	0.06	1.00	0.27	38
Alnus glutinosa	3.77	19.4	194	0.13	0.05	0.51	0.35	27
Prunus avium	3.32	19.9	167	0.12	0.04	0.31	0.32	62
Castanea sativa	3.21	14.3	224	0.10	0.02	0.77	0.24	15
Pinus nigra subsp. nigra	3.08	15.3	202	0.07	0.02	0.15	0.36	15
Tilia cordata	2.58	12.9	199	0.09	0.05	0.12	0.59	29
Quercus rubra	2.09	10.3	202	0.12	0.01	0.09	0.06	8
Acer platanoides	1.66	9.10	183	0.06	0.01	0.00	0.12	18
Robinia pseudoacacia	1.66	9.57	174	0.06	0.02	1.64	0.33	12

Methodology and data sources

Growing stock

Volume over bark of the stems of living trees (diameter at 1.30 m above 7.5 cm, up to a 7 cm top), measured on the inventory plots at their first visit. Plot values per hectare (sum of tree volume × weight) are averaged by department and multiplied by the forest area of the department.

Forest area

Area of the forest classes of the land cover map (broadleaved, coniferous and mixed forest), assigned to departments by the centroid of each land cover polygon. The area of a species is the forest area multiplied by the share of the species in the basal area.

Biological growth

Annual volume increment of the living trees, derived from the radial increment of the last five years (core samples), assuming that tree volume scales with diameter to a species-specific power estimated from the data. Trees without a measured increment receive the mean relative growth of their campaign, species and diameter class. The recruitment of new trees is not included.

Harvest and mortality

Since 2015 every plot is revisited five years after its first visit and the fate of each tree is recorded. The volume at the first visit of the trees found cut (harvest) or dead, broken or windthrown (natural mortality) is divided by five to obtain an annual flow, attributed to the year of the second visit. The mortality rate is the mortality volume divided by the growing stock.

Regions and country

Department totals are summed and ratios are recomputed from the sums. Single-year indicators rest on a small sample of plots and carry a sampling error, larger for small areas, minor species and flows; the `window` option of `process_data` pools consecutive campaigns.

Growth models

Chapman-Richards curves V(age) = a (1 − exp(−b·age))^c are fitted by nonlinear least squares on the living trees whose age at 1.30 m was measured. When a subregion has fewer than 30 aged trees of a species, the model of the enclosing region or of the country is shown.

Variability

A variable is flagged as highly variable when its mean absolute year-on-year change exceeds 5 % of its mean level; for those variables the report compares the year with the previous year and with the mean of the five previous years.

Data sources

- IGN – Inventaire forestier national français, Données brutes, Campagnes annuelles 2005 et suivantes, https://inventaire-forestier.ign.fr/dataIFN/ (Licence Ouverte Etalab 2.0)
- Corine Land Cover 2006, © European Union, Copernicus Land Monitoring Service, European Environment Agency
- Administrative borders: france-geojson (G. David), derived from IGN ADMIN EXPRESS, https://github.com/gregoiredavid/france-geojson